AI-Powered Community Health Request & Fulfillment System

Generative AI Hackathon 2026 Solution Brief

1. Executive Overview

Community Health currently relies on a fragmented and manual workflow, Microsoft Forms submissions routed to email, followed by manual interpretation and spreadsheet tracking. This approach creates inefficiencies, delays, and inconsistent data handling.

Our solution introduces a fully integrated, AI-powered digital platform that transforms this workflow into a structured, intelligent, and scalable system.

The system replaces:

> Microsoft Form -> Email -> Manual Sorting -> Excel

With:

> Smart Intake -> AI Processing -> Central Queue -> Admin Decision -> Automated Routing -> Real-Time Tracking -> Predictive Analytics

This platform significantly improves:

- Operational efficiency
- Data accuracy and consistency
- Decision-making speed
- Visibility into demand, staffing, and geographic coverage

2. System Architecture & Workflow

Our system is designed as a modular, end-to-end pipeline, consisting of nine interconnected phases:

[Visual: Homepage]

2.1 Request Intake Layer (User Interface)

Users (community partners or internal staff) access a modern web interface with multiple intelligent input options:

- Smart Form Interface
- Conversational AI Chat Intake
- Voice-Based AI Intake (High-Impact Feature)

The system captures structured and validated data, including:

- Required fields (name, contact, event details, location, request type)
- Optional fields (attendees, materials, notes, special requirements)

Key enhancements:

- Dropdown-based inputs to reduce ambiguity
- Real-time validation (email/phone regex, valid ZIP codes, future dates)
- AI-assisted suggestions for materials and event types

Conversational & Voice AI Intake (WOW Feature)

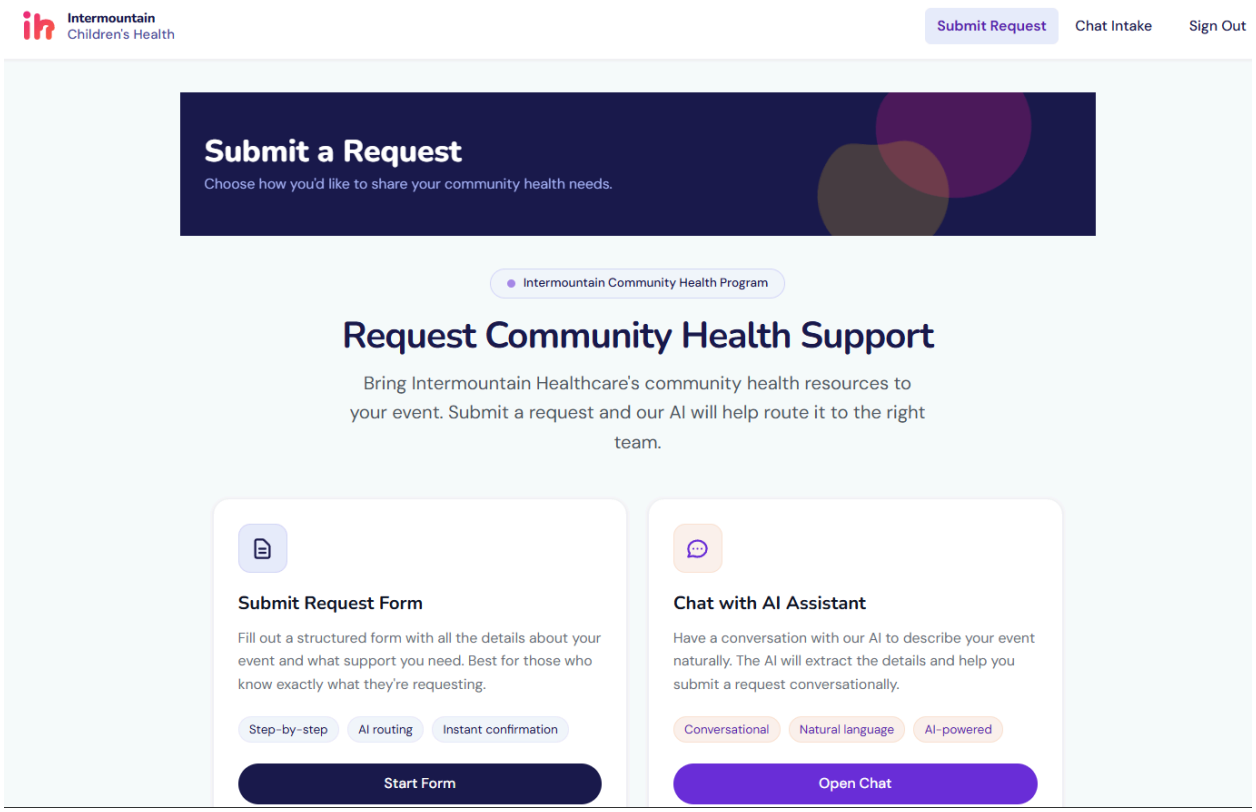
Users can interact naturally via text or voice, similar to a virtual assistant:

> "We're hosting a school health fair with 200 kids"

The AI automatically extracts:

- Event type
- Audience size
- Location
- Recommended materials

This eliminates friction, reduces form fatigue, and improves data quality, especially for non-technical users.



[Visual: User Interface - Smart Form]

2.2 AI Processing Engine

Once submitted, each request is transformed into a structured data object and processed through multiple AI layers:

a. Classification

Automatically determines fulfillment type:

- Staff-supported event
- Mailed materials
- Pickup

The system also cross-validates user selection, flagging inconsistencies.

b. Tagging & Categorization

Assigns:

- Program category
- Material type
- Geographic cluster
- Priority level

c. Natural Language Processing (NLP)

Extracts insights from notes:

- Event type
- Audience characteristics
- Special requirements

d. Error Detection

Flags:

- Missing required fields
- Unrealistic values

e. Recommendation Engine

Suggests:

- Optimal fulfillment pathway
- Staffing requirements
- Priority level

f. Explainable AI

Displays:

- Decision reasoning
- Confidence score
- Key influencing factors

2.3 Centralized Queue System

All processed requests enter a real-time centralized queue, functioning as the operational core of the system.

Each request becomes a trackable ticket with:

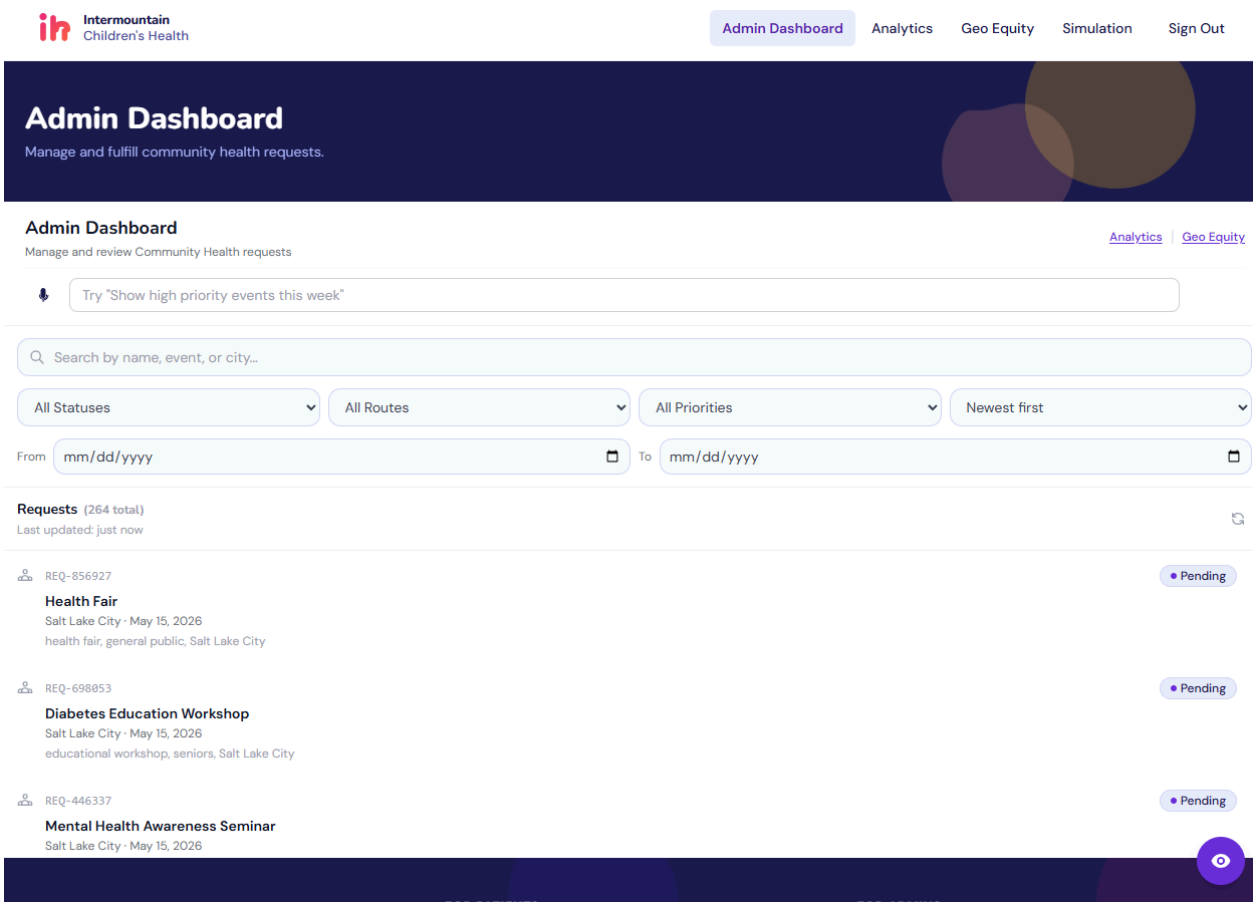
- Event details
- AI-generated tags
- Priority level
- Status

Admins can:

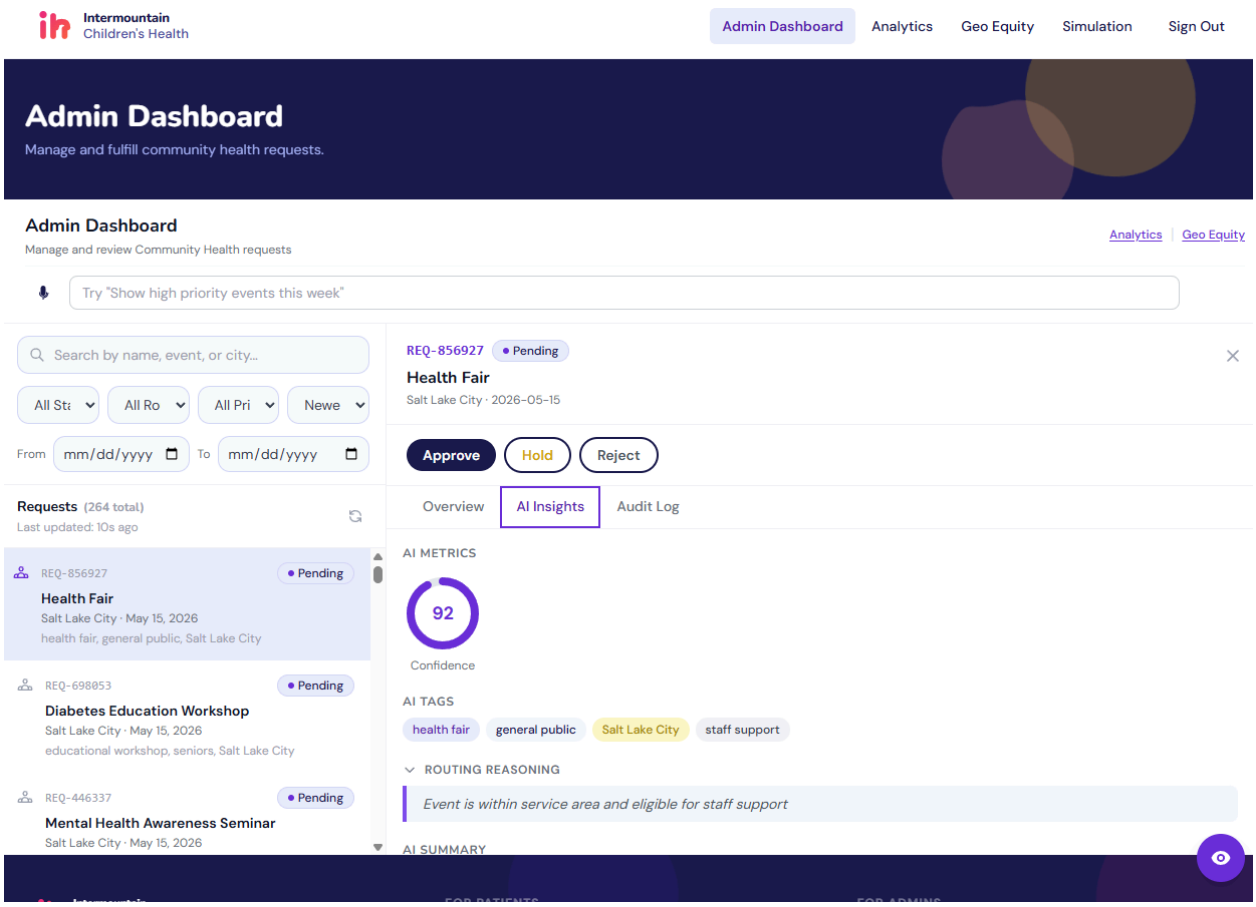
- Filter by date, location, priority, fulfillment type, program
- Search by event name or requestor

Advanced Feature: AI-Powered Querying
Admins can use natural language:

- "Show urgent events this week"
- "Show underserved regions"



[Visual: Queue Dashboard Demo]



[Visual: Queue Dashboard Request View]

2.4 Admin Review & Decision System (Human-in-the-Loop)

The system ensures human oversight while reducing manual effort.

Admins can:

- Review full request details
- View AI insights and recommendations
- Edit or override AI decisions
- Approve, reject, or hold requests

AI Copilot Assistant (Advanced Feature)
Admins can ask:

- "What should I prioritize?"
- "Where is demand highest?"
- "Summarize this week's requests"

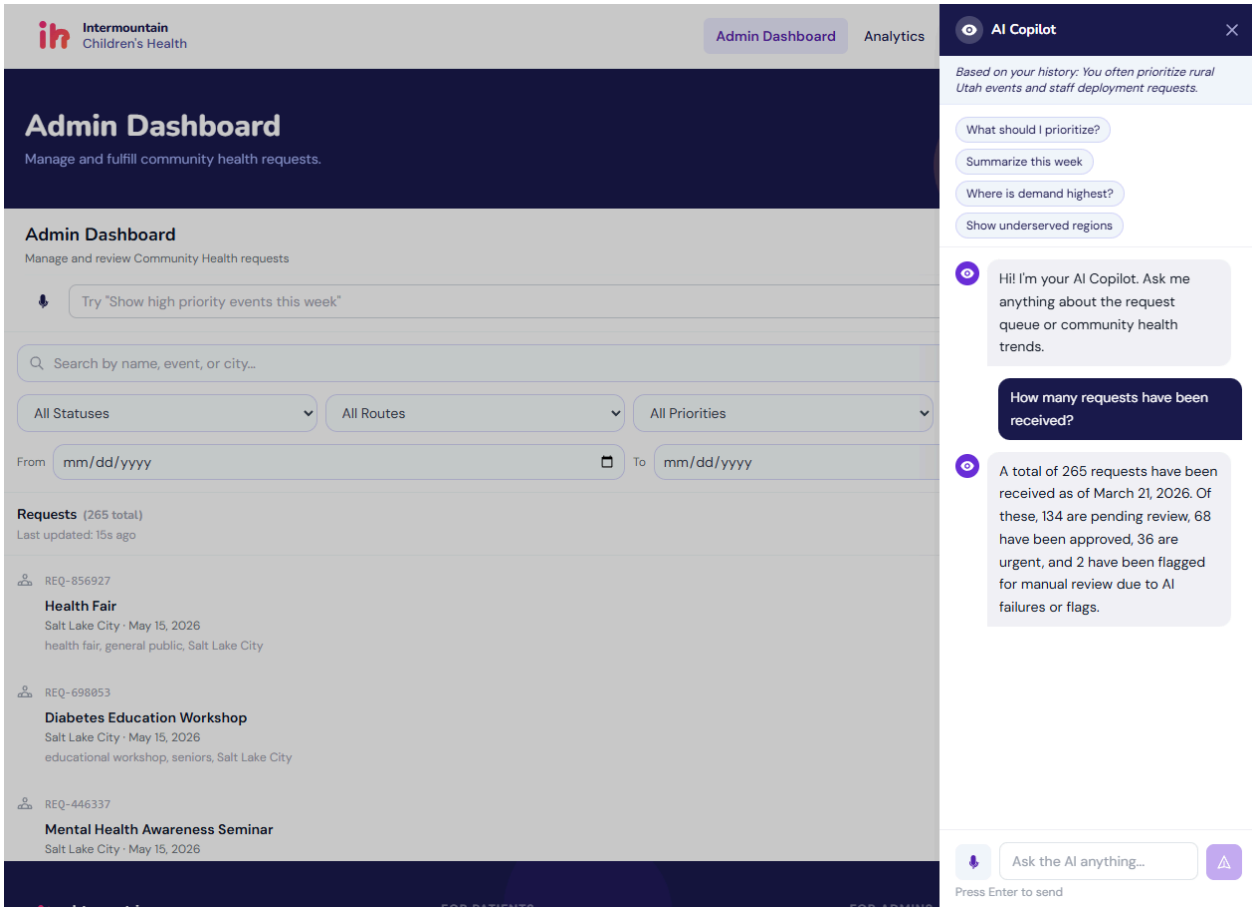
Voice-Enabled AI Copilot

The AI Copilot also supports voice interaction, enabling hands-free decision-making and faster workflow navigation.

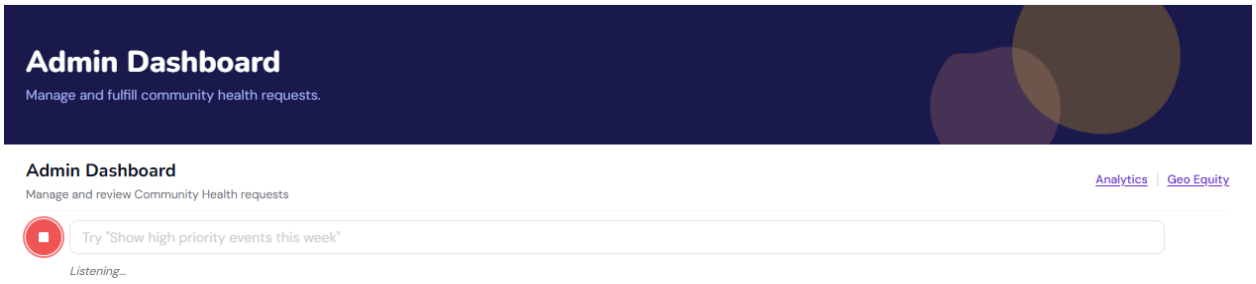
AI Memory System (Advanced Feature)

The system learns from past decisions and provides contextual suggestions:

> "You typically prioritize rural outreach events."



[Visual: Admin Panel with AI Copilot]



[Visual: Admin Panel with Voice Input]

2.5 Fulfillment Logic Engine

The system automates routing decisions based on structured rules and AI insights:

Geographic Logic

- Inside service area -> Staff deployment
- Outside service area -> Mail fulfillment

Priority Logic

- Large events -> High priority
- Near-term events -> Urgent

Supported Fulfillment Types

- Staff-supported events
- Mailed materials
- Pickup

2.6 Approval & System Handoff

Once approved:

- Requests are routed to an external system
- Stored for tracking and reporting
- Trigger automated workflows

Automated Notifications

- Confirmation emails
- Instructions
- Reminder notifications
- Follow-ups and feedback surveys

2.7 Status Tracking System

Each request progresses through a transparent lifecycle:

- Submitted
- In Review
- Approved

- Sent
- Fulfilled

Status is visible across:

- Queue
- Request details
- Dashboard

2.8 Dashboard & Analytics Layer

The system provides real-time, actionable insights:

Key Metrics

- Total requests
- Requests by status
- Fulfillment type distribution

Visual Analytics

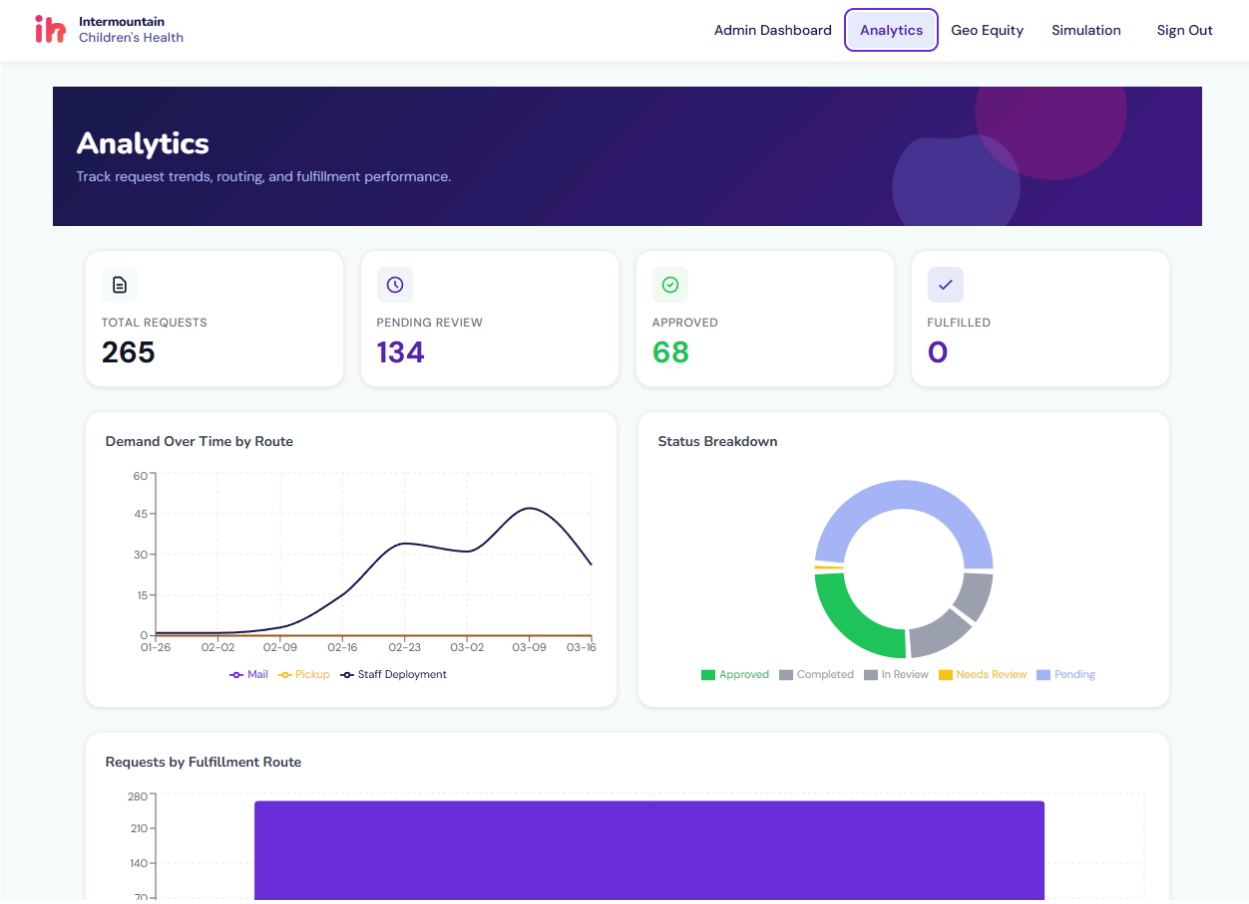
- Demand over time
- Program usage trends
- Staffing requirements

Geographic Intelligence

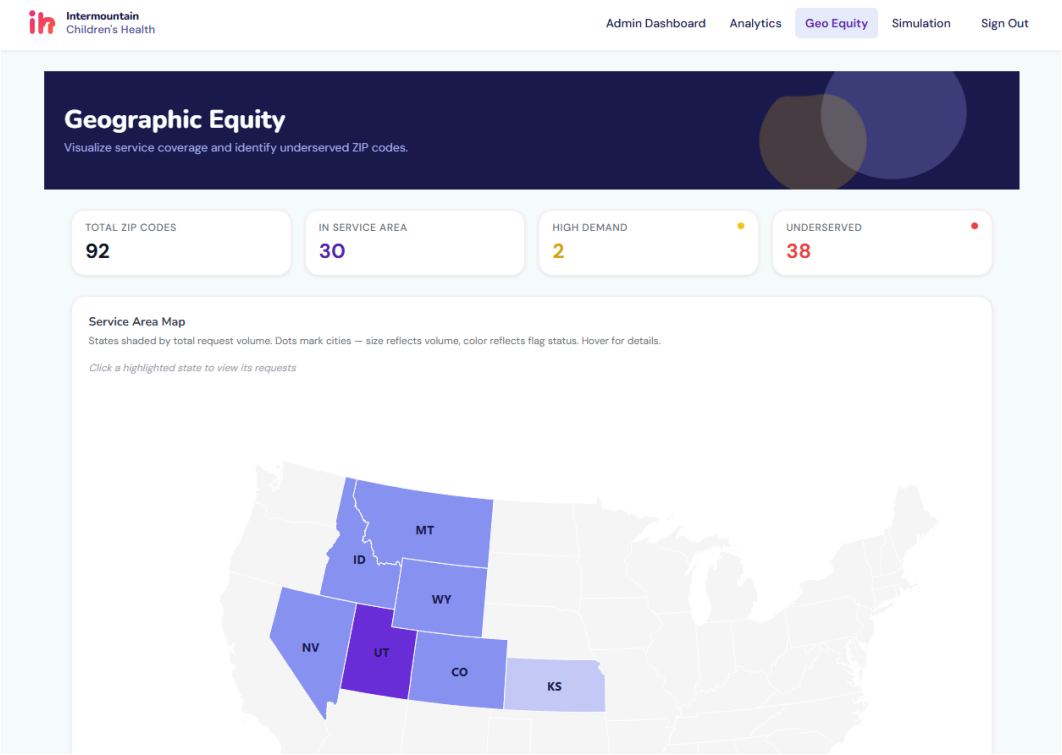
- ZIP-level heatmaps
- Demand distribution maps
- Geo-equity analysis

AI Insights

- Underserved regions
- High-demand areas
- Staffing gaps



[Visual: Analytics Dashboard]



[Visual: Geo Heatmap]

2.9 Advanced AI Features (Innovation Layer)

Our system incorporates several forward-thinking capabilities:

Predictive Analytics

- Forecasts demand trends
- Predicts staffing shortages

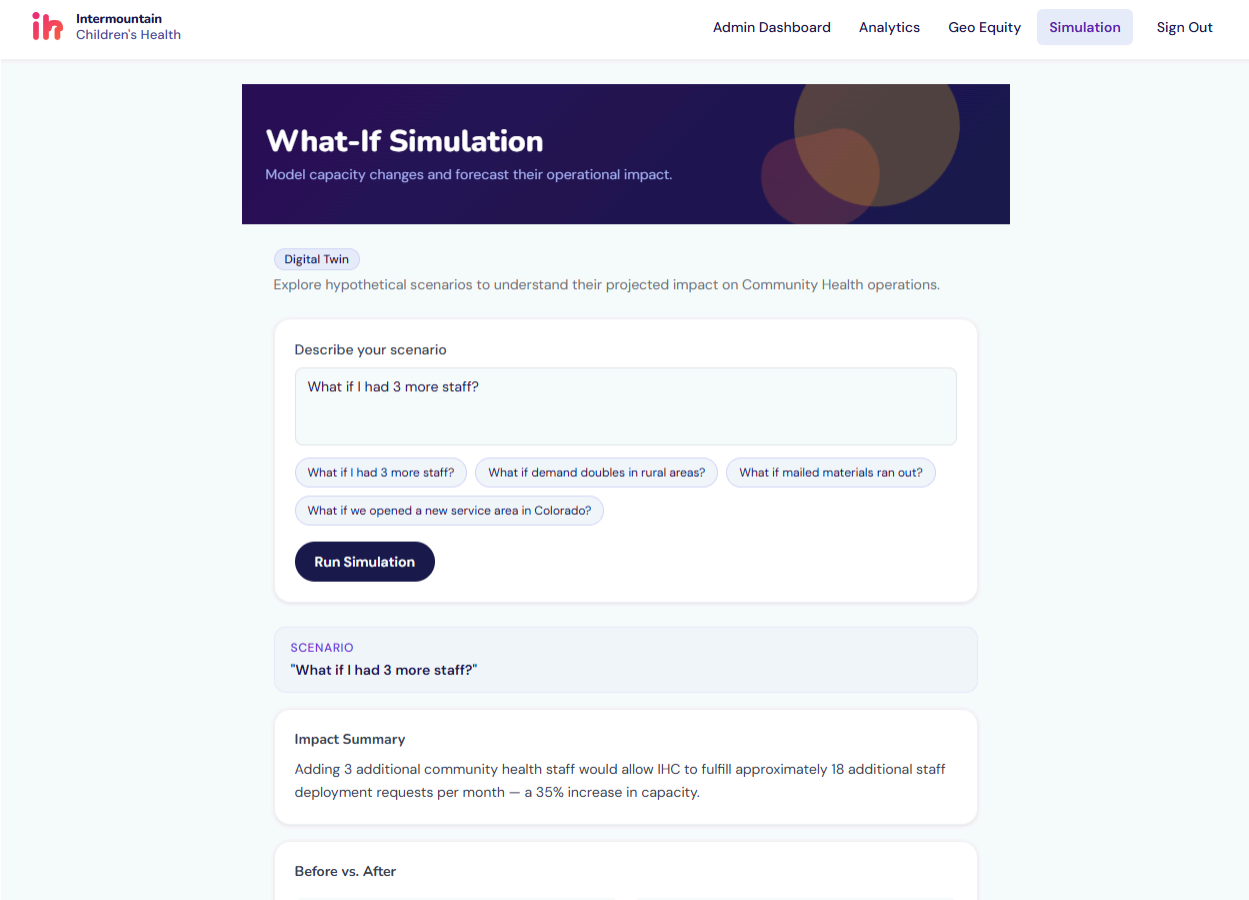
Simulation Mode (Digital Twin)

Admins can simulate scenarios:

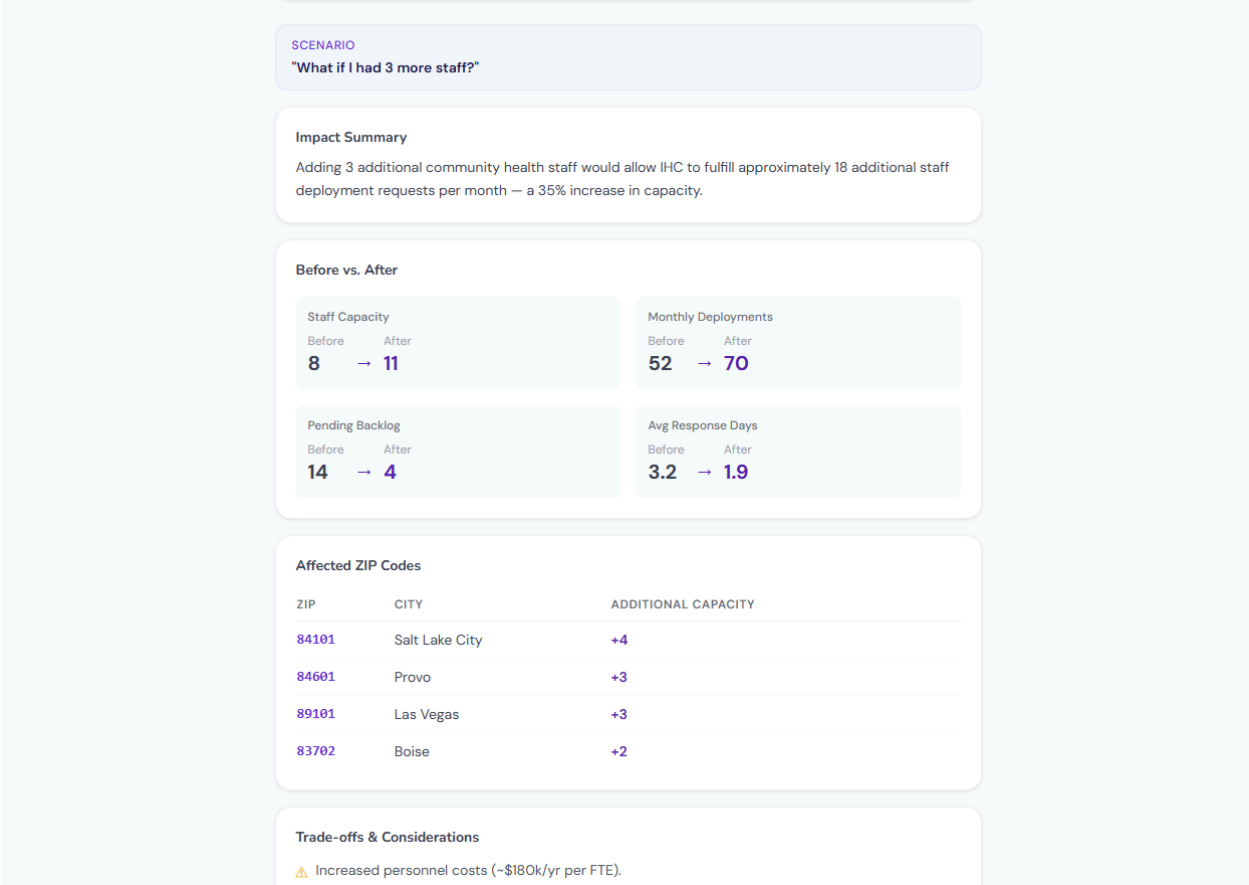
> "What if we increase staffing in rural areas?"

System outputs:

- Coverage changes
- Demand fulfillment improvements
- Resource trade-offs



[Visual: Simulation Mode Interface]



[Visual: Simulation Mode Interface with Dynamic Graphs]

3. AI & Automation Strategy

Our solution integrates AI at multiple levels:

- **Input Intelligence** -> Smart validation and conversational intake
- **Processing Intelligence** -> Classification, tagging, NLP extraction
- **Decision Intelligence** -> Recommendations and prioritization
- **Operational Intelligence** -> Queue automation and routing
- **Strategic Intelligence** ->Predictive analytics and simulation

This layered approach ensures:

- Accuracy
- Scalability
- Explainability
- Human oversight

4. Security & Privacy Considerations

- Use of anonymized/test data
- Input validation to prevent malformed entries (prevent injections)
- Role-based access (admin vs user)
- Secure authentication for admin access
- Compliance with healthcare data handling best practices

5. Scalability & Future Enhancements

The system is designed for real-world deployment and can be extended to:

- Inventory management integration
- Automated staffing calendars
- Real-time resource allocation
- Can do deeper integration with enterprise platforms (e.g., Qualtrics)
- Expansion to additional Community Health workflows

6. Innovation Highlights

Key differentiators of our solution:

- Conversational AI intake with voice capability
- AI Copilot for administrative decision-making
- Natural language interaction with system data
- Simulation-based planning (Digital Twin)
- Organizational memory and adaptive learning
- Latent demand detection for equitable outreach

7. Conclusion

This AI-powered system transforms a manual, fragmented workflow into a smart, scalable, and data-driven platform.

By combining:

- Structured data collection
- Intelligent automation
- Human-in-the-loop decision making
- Advanced predictive capabilities

Our solution enables Community Health to:

- Operate more efficiently
- Serve communities more equitably
- Make proactive, data-informed decisions